

Arya Darshan Roi Vijai

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EDUCATION

Master of Science in Urban Data Science

New York University, Tandon School of Engineering, New York City; May'27; GPA: 4.0 / 4.0, Semester 1, Dec 2025

Bachelor of Technology in Electronics and Communications Engineering

PES University; Oct'23

WORK EXPERIENCE

New York University: *Data Analyst*

Jan'26-present

- Created three impactful data visualizations about NYU Arts & Science center for potential donors using Tableau
- Data extraction, cleaning and analysis to identify trends & patterns from previous newsletters
- Created Data insights led to increased donations, grants and visiting scholars in spring 2026
- Used Adobe Enterprise Manager for managing content across organization

New York University: *Graduate Researcher*

Sep'25-present

- Use of Multimodal LLMs for image-text generation from satellite image data of WUI regions
- Language Model Evaluation
- Ongoing research and potential publication on structural damage during wildfires
- Leveraged NYU HPC resource for running LLaVa 8B params model and other LLMs
- Risk mitigation under natural disasters

SIEMENS: *Software Engineer - Research*

Jun'23-Aug'25

- Developed and implemented an innovative image processing technology to improve Quality Control process in manufacturing (Authored & Published)
- Built novel graph network in Neo4j connecting multiple databases and integrated with LLM agents (agentic AI) for inferencing multiple databases which cut down database query time significantly
- Worked on grid automation workflows by AI creating chatbots for SIEMENS ElectrificationX project
- Used Anthropic model context protocol to automate database workflow, reducing considerable time compared to LLM
- Built data visualizations for software team productivity and presented to Siemens CEO and leadership, influencing company-wide adoption of GitHub Copilot
- Developed software to perform video search chatbot using CLIP technology
- Created a chatbot as a service agent for a product using LLM and RAG
- Automated image annotation and classification using pre-trained YOLO v8 model – cutting down 80% labeling time, leading to increased data scientists' productivity
- Completed and deployed industrial safety project using multimodal LLM and ML
- Employed Docker for running YOLO model inside containers and transmitting inferences externally through API

SIEMENS: *Technical Intern*

Feb'23-Jun'23

- Used linear regression to predict prices of Electric Cars based on specifications and features
- Developed mobile application using React-Native for an OEM (original equipment manufacturer) to monitor OEE (overall equipment efficiency) to enable product compatibility with Industry 4.0 requirements

SIEMENS: *Research Intern*

Aug'22-Oct'22

- Developed streamlined motor control techniques through the implementation of PWM (Pulse Width Modulation) signals, enhancing DC motor operational efficiency and performance
- Created device drivers for Texas Instruments and Microchip microcontroller units

PUBLICATIONS

1. Roi Vijai, Arya Darshan, Kaundinya, S. R., Raghunandan, K. S., & Lohith, M. V. (2025). *Quality check efficiency enhancement through accelerated surface analysis*. Prior Art Publishing GmbH.

2. A. Subramaniam, A. S. Rao, A. Vajrapu, A. D. R. V. and V. H.R, "Real Time Monitoring of Forest Fires and Wildfire Spread Prediction," *2023 2nd International Conference on Applied Artificial Intelligence and Computing (ICAAIC)*, Salem, India, 2023, pp. 772-779, doi: 10.1109/ICAAIC56838.2023.10141168.

PROJECTS

1. **Title:** Mediterranean Futures Summer School – Mar Menor Living Lab (Cartegena, Spain) – Selected for Upcoming Summer Exchange
Duration: Jul 13-17, 2026
Description: Engage with research on complex socio-ecological system where environmental processes interact with economic activities, governance frameworks and community dynamics.
2. **Title :** AI-Powered Real Estate Tax Assessment & Homeowner Representation
Duration: Nov’25 – Dec’25
Description: NYU Stern Data Visualization project to assess and pitch the feasibility of AI in real estate tax assessment for Loeb.nyc
3. **Title:** Real Time Monitoring of Forest Fires and Wildfire Spread Prediction
Duration: Aug’22 – Aug’23
Description: Detection of forest fires their direction and rate of spread using satellites and sensors for real-time detection and tracking, aiding quick response and predicting spread.
4. **Title :** Joint Radar Communication
Duration: Jan’22 – Aug’22
Description: The potential of radar and communication on a single platform was employed in vehicle-to-vehicle communication in autonomous cars, enabling lower interferences between the two.
5. **Title:** User Engagement Patterns on E-commerce Platforms
Duration: Nov’21 – Dec’21
Description: This project utilizes the Pandas library to visualize diverse consumer behaviors on e-commerce sites and discern product viewing intentions.

TECHNICAL SKILLS	LEVEL OF EXPERTISE
Python, ML, Data Analysis, Data Cleaning, Data Extraction, LLM, Databases, Agentic AI, MCP, Retrieval Augmented Generation, Model finetuning, Image Processing, Computer Vision, Data Governance, AWS Bedrock, AWS AgentCore	Advanced
Visualization: Tableau, Python libraries & think-cell	Advanced
Algorithms: Linear Regression, K-means Clustering, SVM, Random Forest, Naïve Bayes, Convolutional Neural Networks, YOLO, ResNet, MobileNet	Advanced
Software Deployment: Docker, Git, FastAPI, Mosquitto MQTT, AWS S3, SQL, AWS EC2, AWS Strands Agents	Intermediate
UI development: Streamlit, React-native, React, Xcode, Adobe Enterprise Manager	Intermediate

AWARDS & CERTIFICATIONS

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| 1. SIEMENS, Sep’24: | Rising Inventor Certificate “IP Samman” award for early publication |
| 2. Coursera, Jan’24: | IBM AI Engineering Professional Certification |
| 3. SIEMENS, Dec’23: | AI Hackathon Finalist |
| 4. PES University, Aug’20: | Earned the Distinction Award for Engineering Academic Excellence |

HACKATHONS & PRESENTATIONS

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| 1. NYU, 2026: | Participated in two hackathons leveraging data science for Urban Futures
Presented Geospatial data using tableau in Re/presenting data |
| 2. SIEMENS, 2024: | Presentation on chatbot projects during the ‘MLXPO Series 1 & 2’ |
| 3. SIEMENS, 2023: | <ol style="list-style-type: none"> a. ‘AWS DeepRacer’ for finetuning ML model params for reinforcement learning b. SIEMENS Machine Learning hackathon |
| 4. PES University, 2023: | Presented paper on ‘Real Time Monitoring of Forest Fires and Wildfire Spread Prediction’ at the 2nd International Conference on Applied Artificial Intelligence and Computing. |
| 5. PES University, 2020-21: | Completed two terms with the Centre for Innovation and Entrepreneurship, presenting for workshops on design thinking & agile practices |

EXTRACURRICULAR ACTIVITIES

SIEMENS, Bangalore, 2024:	Organized MLXPO (Machine Learning Expo) event
SIEMENS, Bangalore, 2024:	Participated in SIEMENS Cricket & TT Tournaments
NGO for the Blind, Jun'18:	Volunteered as tutor for visually impaired children